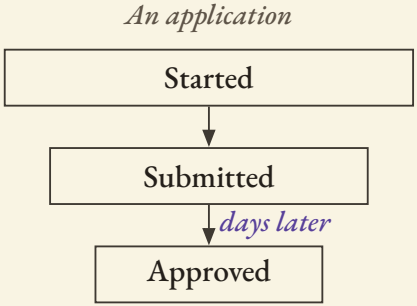
Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.



John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

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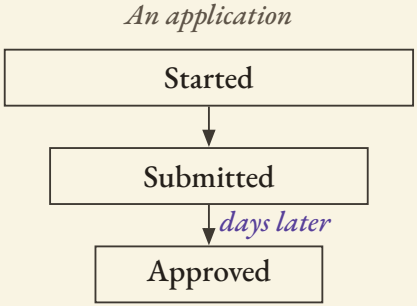
A page that wins the first step can lose the last.



John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*
Pair A	6	3	1
Pair B	4	4	4
Pair C	3	2	3

The first step alone would choose pair A.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*	*Sum*
Pair A	6	3	1	10
Pair B	4	4	4	12
Pair C	3	2	3	8

The first step alone would choose pair A.
Amazon shows the highest sum: pair B.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers

Approvals rose by a high single-digit percentage.
relative change so far, not reported as significant

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

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Amazon: “Thorough exploration can confirm that a content pool contains no winner.”

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.